

Microwave absorption properties of $\text{Ni}_{0.6}\text{Zn}_{0.4}\text{Fe}_2\text{O}_4$ composites with nonmagnetic nanoferrite percentages

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ABSTRACT

Microwave absorption materials need to be thin and lightweight and possess strong wave absorption ability and a wide absorption frequency band. To satisfy these conditions, we changed the microstructure of the composite by mixing ferrite material with different particle sizes. Specifically, we mixed nanosized nonmagnetic ZnFe_2O_4 powder into $\text{Ni}_{0.6}\text{Zn}_{0.4}\text{Fe}_2\text{O}_4$ powder, investigated the microwave absorption properties depending on the packing fraction. The crystal structure of the synthesized ferrite powders was analyzed through XRD, and the particle size was analyzed using a PSA and SEM. The density of the powders, which is required to measure the packing fraction, was determined via the gas disposition method, and the magnetic properties of the composites were analyzed using a VSM. The reflection loss represents the electromagnetic wave absorption characteristics, and it was calculated by substituting the measured permittivity and permeability values, into the equation based on the transmission line theory. The $\text{Ni}_{0.6}\text{Zn}_{0.4}\text{Fe}_2\text{O}_4/\text{ZnFe}_2\text{O}_4$ composite showed 99.9% absorption with a high packing fraction, and the absorption peak shifted to high frequencies. These characteristics suggest that the absorption ability and frequency range of the electromagnetic-wave-shielding composite can be easily controlled. Because of the high-absorption characteristic, absorption frequency control, and cost effectiveness, this composite can be applied to products such as thin electromagnetic-wave-shielding sheets.

1. Introduction

Although electronic devices employing electromagnetic (EM) waves make our lives convenient, they are accompanied by EM radiation hazards. Since the EM spectrum is becoming increasingly crowded, EM interference (EMI) may be generated, which interferes with the normal operation of electronic devices [1]. Moreover, it can affect the brain and nervous system in the human body and cause fatigue, headache, dizziness, sleep disturbance, cancer, etc. [2] Via experiment on a mouse, a previous study proved that EM waves cause hippocampal lipidome [3]. Therefore, materials that can appropriately shield EM waves need to be researched. Methods of shielding EM waves include reflection and absorption. Reflected EM waves may affect other circuits inside electronic devices or may adversely affect the human body. Therefore, microwave absorbing materials that absorb EM waves and emit heat energy are more widely used than materials reflecting EM waves [4,5].

Incident microwaves comprise electric and magnetic fields, and they interact with the microwave absorption materials. Shielding mechanisms can be classified into electric and magnetic interactions [6]. A

representative material with an EM absorption property due to magnetic interaction is ferrite materials. Among them, spinel ferrite materials exhibit high complex permeability and magnetic resistivity in the high frequency band; thus, they are promising for use as EM wave absorbing materials [7–9]. Spinel ferrite (MFe_2O_4) is an iron compound that bonds with metal compounds (M). Mn, Zn, Mg, Cu, Ni, or Co is substituted at the M-site, yielding Ni–Zn ferrite [10], Ni ferrite [11], Ni–Cu–Zn ferrite [12], etc. Ferrites are synthesized via a solid-state reaction method [13], sol-gel method [10], hydrothermal method [14], or coprecipitation method [15], and they exhibit different magnetic characteristics according to the chemical composition [16], sintering temperature [17], etc. Magnetic saturation is a parameter that has the greatest influence on the EM wave shielding performance and requires a high sintering temperature. However, the volume contracts after the high-temperature sintering, and for Ni–Zn ferrites, it shrinks by up to 20%, which makes its use in applications difficult [18]. Thus, a ferrite that is used as an absorber is mixed with a binder and employed. The types of binders are epoxy resin [12], wax [19], paraffin [20], etc. The EM shielding effect of these polymer composites has been shown for the sample thickness of

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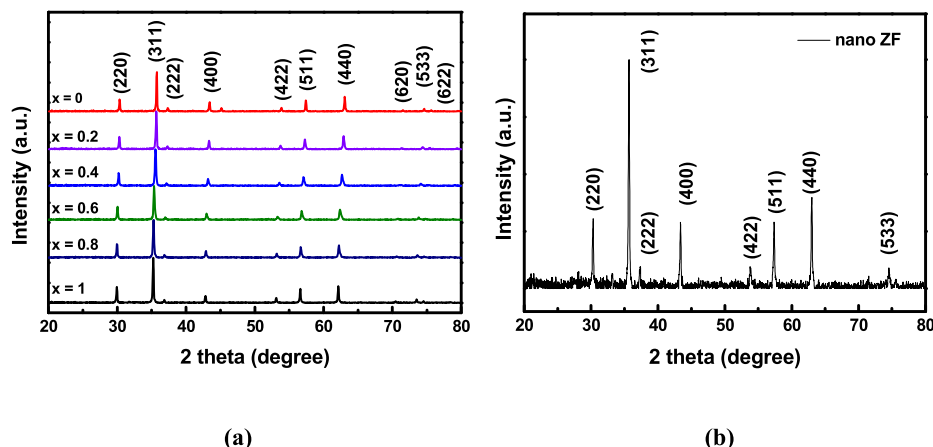


Fig. 1. (a) XRD patterns of $\text{Ni}_{1-x}\text{Zn}_x\text{Fe}_2\text{O}_4$ by solid-state reaction method, and (b) ZF by coprecipitation method.

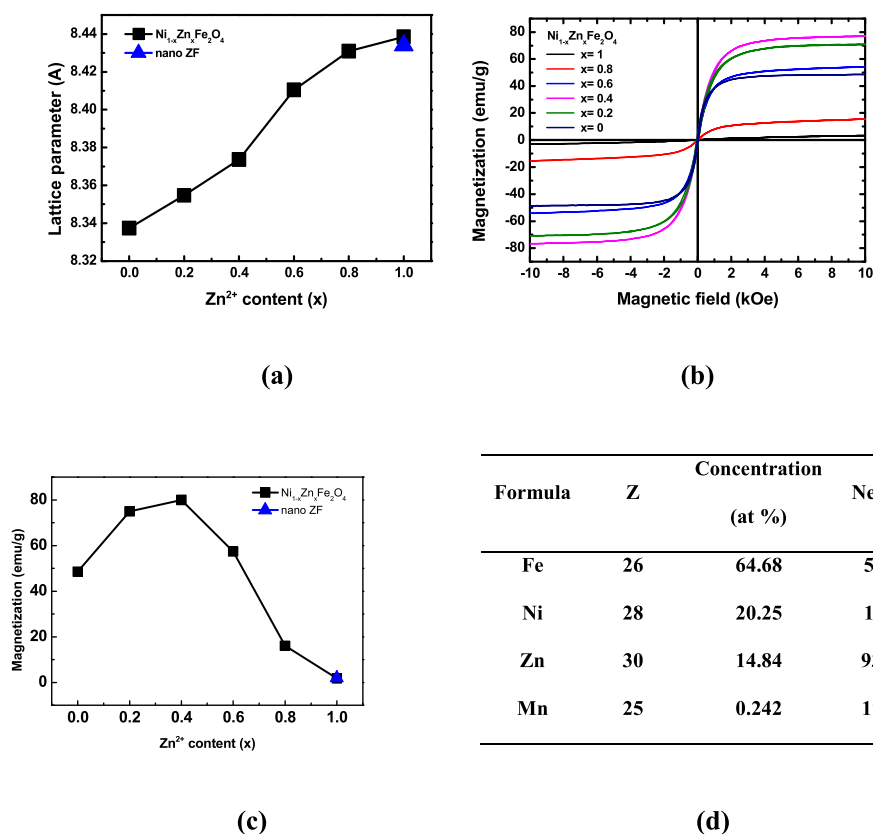


Fig. 2. (a) Lattice parameter and (c) magnetic saturation properties of the synthesized ferrite powders. (b) Hysteresis loop of $\text{Ni}_{1-x}\text{Zn}_x\text{Fe}_2\text{O}_4$ prepared using the solid-state reaction method, and (d) XRF analysis of the $\text{Ni}_{0.6}\text{Zn}_{0.4}\text{Fe}_2\text{O}_4$ powder composition.

Table 1

Effect of Zn^{2+} content(x) on $\text{Ni}_{1-x}\text{Zn}_x\text{Fe}_2\text{O}_4$ parameters estimated from XRD and VSM.

x	a (Å)	Vol (Å ³)	M _s (emu/g)	Coercivity (G)
0.0	8.338	590.1	48.563	43.093
0.2	8.354	583.0	74.915	48.846
0.4	8.369	586.1	79.990	33.229
0.6	8.410	594.8	57.383	17.891
0.8	8.430	599.0	16.056	14.628
1.0	8.438	600.7	1.804	21.757

about 1.5–9 mm. Microwave absorption materials require thin, light, and strong wave absorption availability as well as wide absorption frequency bands. Furthermore, absorption properties can be improved by adding conductive materials, such as graphene [21], graphite [22], MXene [23], and polyaniline [24].

Although studies on composites applying ferrite's magnetic characteristics and EM absorption properties have been reported, the correlation between the absorption characteristics of composites based on sample microstructures has been little studied. Further, the changes in electromagnetic absorption characteristics with the changes in original powder-particle shape [25], as well as the variations in physical characteristics according to changes in assembly of organic nanoparticles have been studied by various authors [26]. However, the packing

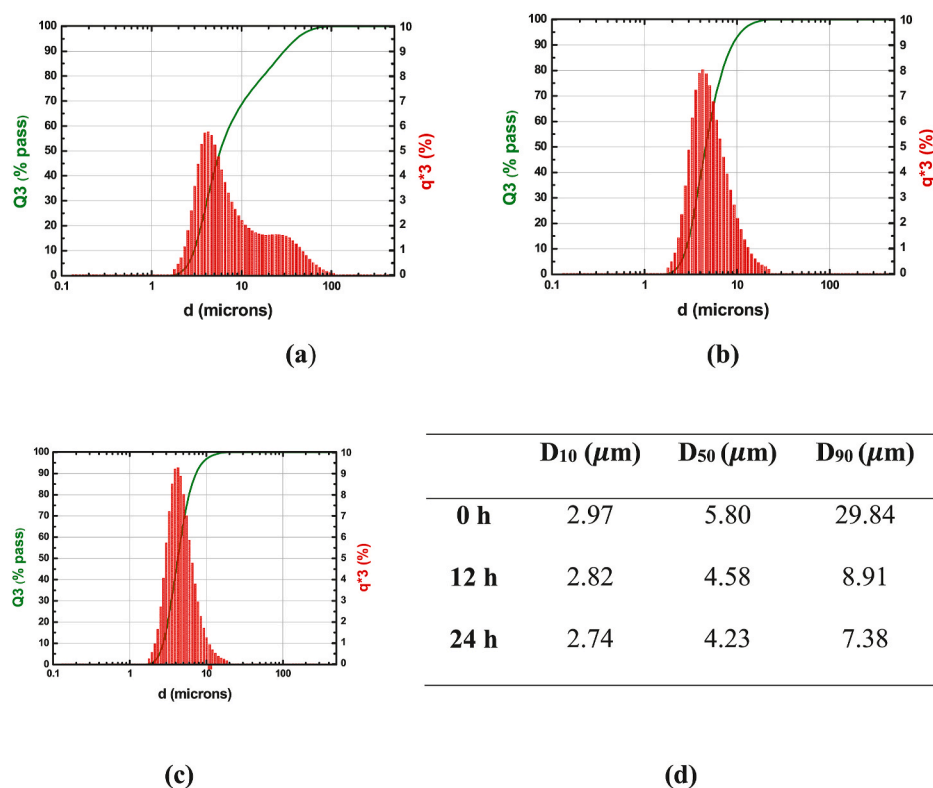


Fig. 3. (a) Particle size distribution of NZF powder via the solid-state reaction method (b) ball-milled for 12 h and (c) 24 h (d) particle size data of NZF powder by milling time.

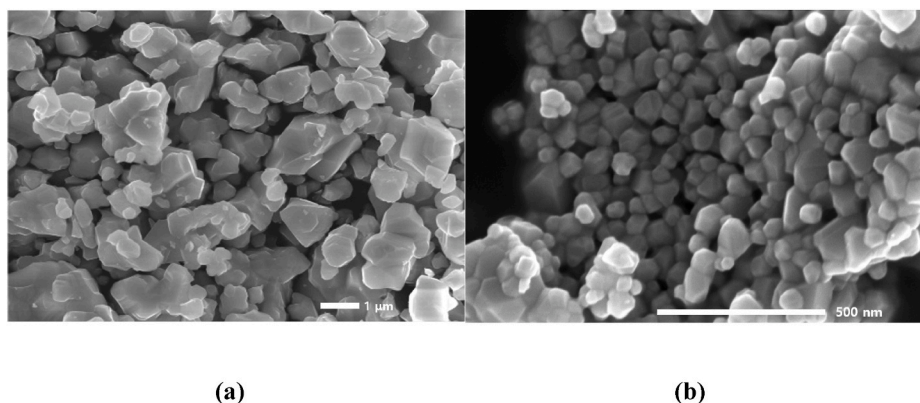


Fig. 4. (a) SEM images of the NZF powder prepared via the solid-state reaction method and ball milled for 24 h, and (b) ZF nanopowder prepared via the coprecipitation method.

fraction-dependent variations in the electromagnetic absorption characteristics remain hitherto unexplored. Herein, the microstructure was changed by changing the packing fraction of the two powders mixed, and two types of spinel ferrites with different chemical compositions and particle sizes were prepared. The base material powder was the micrometer-sized $\text{Ni}_{0.6}\text{Zn}_{0.4}\text{Fe}_2\text{O}_4$ (NZF) that was synthesized via the solid-state reaction method. The small powder was nanosized ferrite powder that was synthesized via the coprecipitation method. The following small ZnFe_2O_4 (ZF) powder was used to analyze the relation between the packing fraction and EM wave shielding characteristics.

The solid-state reaction method is a simple method of heat-treating mixing oxide compounds at the desired ratio, and the coprecipitation method advantageously yields nanosized crystal powders at relatively lower heat treatment temperatures than the solid-state reaction method. The synthesized ferrite powder was manufactured as a composite mixed

with a binder, and it exhibited different magnetic properties and packing fractions depending on the ratio of the nanopowder concentrations. The EM shielding characteristics were studied at the point where the packing fraction was maximized by filling the empty space of the epoxy binder layer with the nanopowder.

2. Experimental details

2.1. Synthesis of $\text{Ni}_{0.6}\text{Zn}_{0.4}\text{Fe}_2\text{O}_4$ by solid-state reaction method

A polycrystalline sample fabricated via the solid-state method was synthesized using high-purity NiO (99.998%, Sigma-Aldrich), ZnO (99.999%, Sigma-Aldrich), and Fe_2O_3 (99.99%, Sigma-Aldrich). The amount of x substitution on $\text{Ni}_{1-x}\text{Zn}_x\text{Fe}_2\text{O}_4$ ($x = 0.2, 0.4, 0.5, 0.6, 0.8$, and 1.0) was calculated using the molar ratio. Based on the x substituted

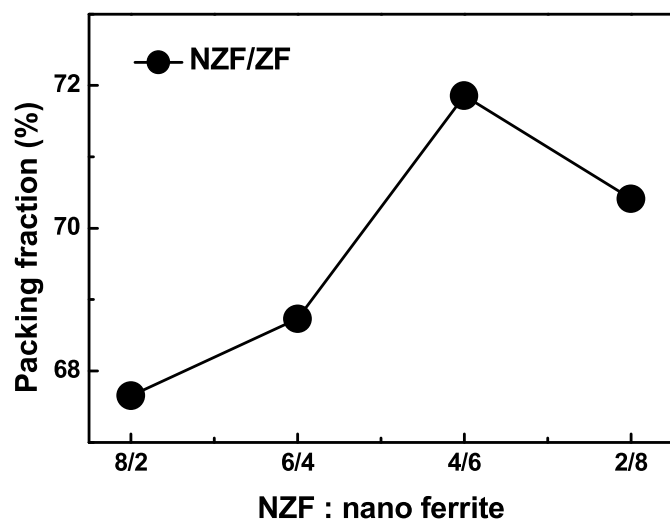


Fig. 5. Packing fraction of the NZF/ZF composite.

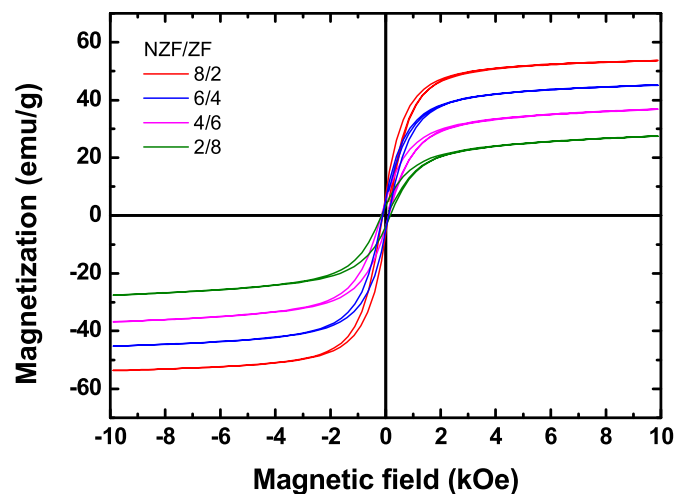


Fig. 6. Hysteresis loop of the NZF/ZF composite.

amount, chemicals were sufficiently mixed and ground using agate mortar and pestle. The mixed powders were pre-annealed at 900 °C for 24 h. The dried powder was mixed and ground again using an agate mortar and pestle. Then, it was pressed at 3 ton/cm² using a circular mold to afford pellets. The as-prepared pellets were sintered at 1,300 °C in tube furnace for 6 h and then ball-milled at 350 rpm for 24 h using zirconia balls. The completed Ni–Zn ferrite powders were dried overnight in a 100 °C dryer.

2.2. Synthesis of ZnFe₂O₄ via the coprecipitation method

ZF nanopowders were synthesized via the coprecipitation method using ZnCl₂ (Sigma-Aldrich) and FeCl₃·6H₂O (Sigma-Aldrich). Solutions A, B were prepared to afford the ZF nanopowder via the coprecipitation method. Solution A was obtained by solvating ZnCl₂ and FeCl₃·6H₂O in 200 mL of deionized (DI) water, and the mixed solutions were stirred for more than 30 min. Solution B was obtained by adding 3 M KOH to maintain strong alkalinity (pH 10.5) and to accelerate the coprecipitation reaction. Solution B was added dropwise at a rate of 2.5 mL/min, and the mixture stirred for 24 h while maintaining a temperature of 80 °C. Then, the obtained sediment was vacuum filtrated, and it was sufficiently washed using DI water and ethanol. The final coprecipitate was dried overnight at 100 °C, and ZF were sintered at 800 °C, for 5 h at

a heating rate of 10 °C/min.

2.3. Preparation of ferrite composites

The NZF/ZF composite sample was physically mixed with the synthesized Ni–Zn ferrite fabricated using the solid-state reaction method and the ZF nanopowders were fabricated via the coprecipitation method at a ratio of 10:0, 8:2, 6:4, 4:6, 2:8, and 0:10. As a binder, a TODAISU product comprising phenyl-based resin 40%–50%, vinyl epoxy resin 26%, H₂O 16%–17%, and EtOH 32%–38% was used. 3 wt% of the binder was added to the mixed powder. The toroidal composite properties for measuring the reflection loss were inner and outer diameters, 3 and 7 mm, respectively. To compare the packing fraction in a toroidal mold at the same pressure, a pressure of 2 ton/cm² was applied to all the composites. The fabricated toroidal samples were dried at 160 °C for 2 h.

2.4. Characterization

The crystal structure of the samples was analyzed with an X-ray diffractometer (XRD, Rigaku Miniflex) using a CuK_α radiation source ($\lambda = 1.54056$ Å). The particle size of the powder synthesized by the solid-state reaction method was measured using a particle size analyzer (PSA, Microtrac S3500), and the Ni_{0.6}Zn_{0.4}Fe₂O₄ powder compositions were confirmed by X-ray fluorescence (XRF) measurements. The size of the ZF nanopowder synthesized via the coprecipitation method was calculated using the Scherrer formula and by substituting the XRD data parameters. The size and morphology of all the powders were observed using a scanning electron microscope (SEM, JEOL JSM-7610F PLUS). The packing fraction of the composites was calculated using Equation (2), while the density required for calculation was measured by a gas pycnometer (AccurPyc 1340). The DC magnetic characteristics were measured by a vibrating sample magnetometer (VSM, Lakeshore model 7404). EM parameters were measured using a PNA-X network analyzer (Agilent Technologies, N5245A) with a frequency range of 0–18 GHz. The thicknesses of the samples were 2.0 mm, and all samples were measured at room temperature.

3. Results and discussion

3.1. Characteristics of Ni_{1-x}Zn_xFe₂O₄

To select the optimal composition of the Ni_{1-x}Zn_xFe₂O₄ powder that is used as the base material, the crystal structure and magnetic characteristics were examined while changing the x value. The substitution amount of x was changed in steps of 0.2. Microwave absorption materials at high frequency, over GHz range, require a high saturation magnetization which is possible to make thinner absorbers [27]. Therefore, we chose the composition with highest saturation magnetization among various Ni_{1-x}Zn_xFe₂O₄ powders. Fig. 1(a) illustrates the XRD pattern of the Ni_{1-x}Zn_xFe₂O₄ ($x = 0, 0.2, 0.4, 0.6, 0.8, \text{ or } 1.0$) powder synthesized via the solid-state reaction method. The XRD pattern confirmed the crystallinity and structure of the powder at 2 theta = 20°–80°. The diffraction peaks of all the synthesized powders were (111), (220), (311), (222), (400), (333), and (440) with cubic phases, and they matched those in ICDD card number #74-2081.

In Fig. 2(a), the lattice parameter increases with the degree of substitution of Zn²⁺ ions, which is due to the difference in the ionic radii (Ni²⁺: 0.8 Å and Zn²⁺: 0.88 Å) [28]. Fig. 2 (b) displays the M – H hysteresis loop for different Ni_{1-x}Zn_xFe₂O₄ powder compositions, and Fig. 2 (c) shows a schematic to easily identify the magnetic saturation properties. With x substitution, the magnetic coercivity gradually decreased, and the magnetic saturation value was the highest at Ni_{1-x}Zn_xFe₂O₄ ($x = 0.4$). The base material of composite was selected as NZF as it exhibited the highest magnetic saturation value of 79 emu/g. Fig. 2 (d) shows the Ni_{0.6}Zn_{0.4}Fe₂O₄ composition confirmed via XRF measurements, indicating that the composite was well synthesized by

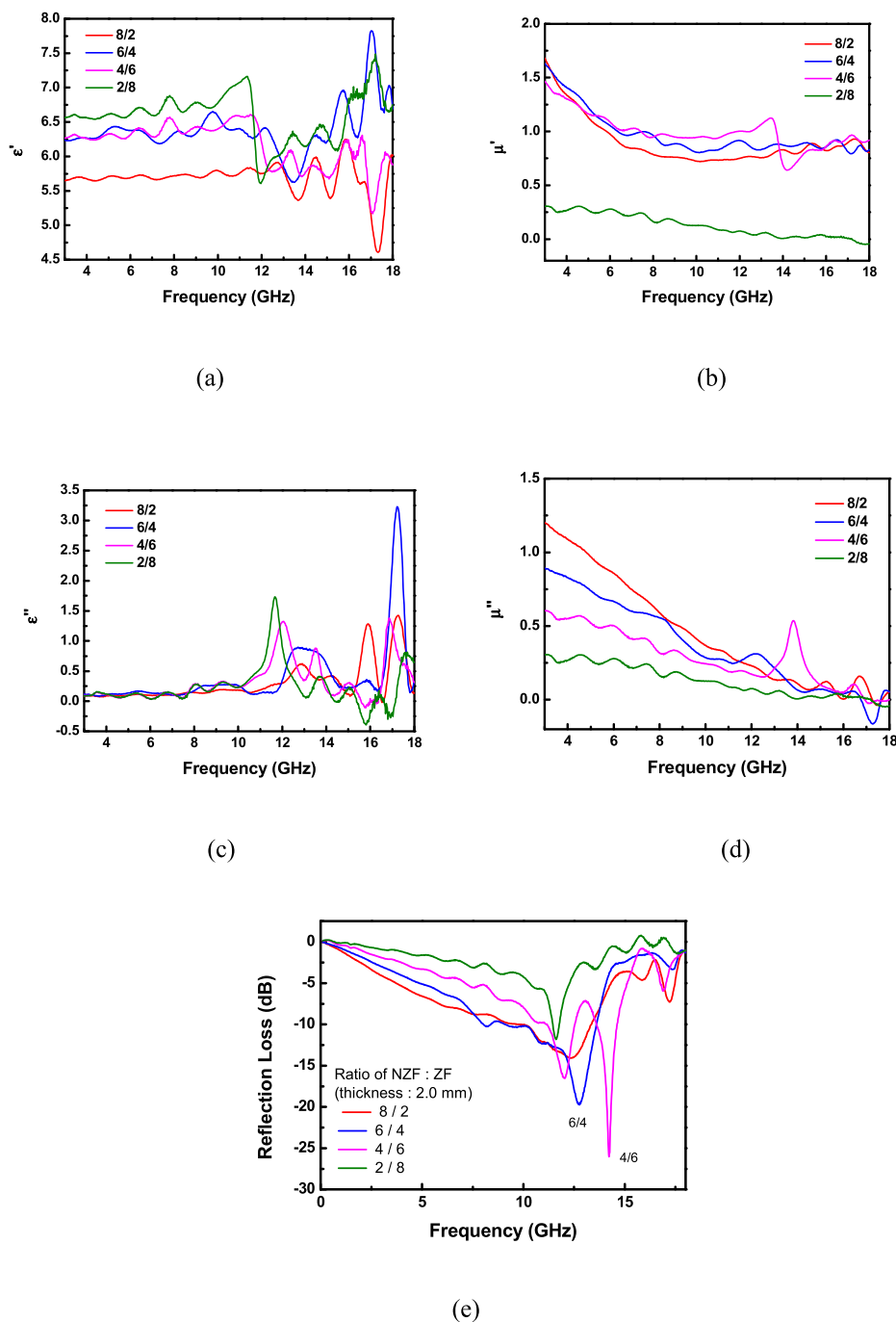


Fig. 7. f dependences of (a) the real and (c) the imaginary parts of the complex permittivity and (b) the real (d) the imaginary parts of the complex permeability of NZF/ZF composites (e) Reflection loss of the NZF/ZF composites at $1 < f < 18$ GHz.

the solid-state reaction method. The effect of Zn^{2+} on the parameters of $\text{Ni}_{1-x}\text{Zn}_x\text{Fe}_2\text{O}_4$ estimated from XRD and VSM is expressed on Table 1.

To accurately analyze the effect of the packing ratio, the selected base material was maintained at a constant average size. Fig. 3 shows the size distribution, analyzed using the PSA, of NZF before and after ball milling. D_{10} , D_{50} , and D_{90} are portions of particles with diameters smaller than 10%, 50%, and 90%, respectively. D_{50} is also known as the median diameter. Before the ball milling, D_{50} , D_{10} , and D_{90} were 5.8, 2.97, and 29.84 μm , respectively, indicating a very wide powder size distribution. Ball milling was performed for 12 and 24 h. After 12 h, D_{10} , D_{50} , and D_{90} were 2.82, 4.58, and 8.91 μm , respectively, and after 24 h, they were 2.74, 4.23, and 7.38 μm , respectively.

The SEM image in Fig. 4(a) shows the NZF morphology after 24 h of

ball milling, and the obtained results were similar to those of D_{10} , D_{50} , and D_{90} analyzed by PSA. As for the NZF used as the base material, the 24-h ball-milled powder with the smallest difference in size distribution was selected.

The packing fraction was determined by employing the nonmagnetic ZF nanopowders as an additive powder. To investigate the correlation between the packing fraction and EM wave shielding property. As represented in Fig. 1(b), the diffraction peaks of the powder were (200), (311), (222), (400), (422), (511), and (440) with cubic phase, and they matched the data from ICDD card number #89-1009. The crystal size was calculated using the strongest peak in the XRD pattern. Additionally, the average crystal size was calculated using the Scherrer formula:

Table 2

Packing fraction, magnetic properties, and effective bandwidth of the NZF/ZF composite.

Sample		Packing fraction (%)	Magnetic saturation (emu/g)	Reflection loss (dB)	Absorption ability (%)
NZF/ ZF	8/2	67.6	53	−14 dB @ 12.4 GHz	<90%
	6/4	68.7	45	−19.7 dB @ 12.7 GHz	<99%
	4/6	71.8	36	−26 dB @ 14.2 GHz	<99.9%
	2/8	70.4	27	−11 dB @ 11.5 GHz	<90%

$$d = \frac{k\lambda}{\beta \cos \theta} \quad (1)$$

Here, d represents the average particle size in nm, k denotes the Scherrer constant ($k = 0.94$), λ is the X-ray wavelength ($\lambda = 1.54056 \text{ \AA}$), β is the X-ray full width at half maximum, and θ denotes the Bragg angle [29]. Using this formula, the crystal size of ZF nanopowders were calculated as 63 nm. Fig. 4(b) displays the SEM images of the prepared powders, and both images well match the average particle size calculated using the Scherrer formula.

3.2. Packing fraction

The density value is required to calculate the packing fraction. The density of the powder was measured via a gas displacement method using helium gas. Density calculations using the gas displacement method are considerably more accurate and reproducible than those using the traditional Archimedes water displacement method. The measured densities of NZF and ZF were 4.8269 and 4.1755 g/cm³, respectively. The composite volume was determined by measuring the radius and thickness and substituting them into the density formula. The two density values were substituted into the packing fraction formula (Eq. (2)). Fig. 5(a) graphically shows the calculated values.

$$\text{packing fraction} = \frac{\text{Core density}}{\text{powder density}} \times 100 \quad (2)$$

For the packing fraction of NZF/ZF composite, the 4/6 ratio composite exhibited the highest packing fraction of 71%, and it decreased for the 2/8, and 0/10 composites.

3.3. Magnetic properties

The M – H hysteresis loop of the composite is shown in Fig. 6 and it varies according to the powder characteristics. The magnetic saturation (Ms) value of the ZF powder is 1.8 emu/g, as evident from Fig. 2 (b) and (c). As the ZF content in the composite was increased, the Ms of the composite decreased, which can be attributed to the decrement in the magnetic component of the composite. The Ms value of the 8/2 composite is 53 emu/g, whereas those of the 6/4, 4/6, and 2/8 composites are 45, 36, and 27 emu/g, respectively.

3.4. Reflection loss

In Fig. 7 (e), the reflection loss (RL) representing EM wave absorption characteristics was calculated and shown based on the complex permittivity and permeability values measured according to the frequency. The RL values were calculated at frequencies in the range of 0–18 GHz for the 8/2, 6/4, 4/6 and 2/8 composites. The 10/0 and 0/10 composites with the absorption ability of less than 90% were excluded from Fig. 7 (e). The absorption ability of RL was calculated using Equations (3) and (4) based on the transmission line theory.

$$\frac{Z_{in}}{Z_0} = \sqrt{\mu_r/\epsilon_r} \tanh \left[j \frac{2\pi f d}{c} \sqrt{\mu_r \times \epsilon_r} \right] \quad (3)$$

$$RL(\text{dB}) = 20 \log \frac{Z_{in}/Z_0 - 1}{Z_{in}/Z_0 + 1} \quad (4)$$

Here, Z_{in} is the incident impedance of the material, Z_0 is the impedance of the free space, ϵ_r is the complex permittivity expressed as $\epsilon' - j\epsilon''$, μ_r is the complex permeability expressed as $\mu' - j\mu''$, f is the frequency of the incident cell wave, c is the speed of the free electron, and d is the thickness of the absorber. In Equation (3), Z_{in}/Z_0 is obtained by substituting the real and imaginary values (i.e., ϵ' , ϵ'' , μ' , and μ''). Additionally, the thickness values of the complex permittivity and permeability rates were obtained through the network analyzer, and RL spectra are obtained by substituting them into Equation (4) [30].

Fig. 7 (a)–(d) show the values of ϵ' , ϵ'' , μ' , and μ'' obtained using the network analyzer. The real parts of ϵ' and μ' represent electric and magnetic storage, while the imaginary parts of ϵ'' and μ'' represent the dielectric and magnetic loss capability of the composite. The value of ϵ' was high in the composite with a high nano ZF content. The increase and decrease of ϵ' was not significant until around 11 GHz; however, after 11 GHz, the ϵ' of the 8/2 and 6/4 composites decreased, whereas that of the 6/4 and 2/8 composites increased. The value of ϵ'' also showed a change in the frequency range after 11 GHz. In particular, the peak of the 2/8 composite shown at 11.5 GHz coincided with the RL peak of the 2/8 composite as shown in Fig. 7 (e). The peak of the 4/6 composite shown at 12 GHz coincided with the second RL peak of the 4/6 composite as shown in Fig. 7 (e). Other composites also showed large ϵ'' values, but the RL peak intensity was very weak. The value of μ' was the largest for the 4/6 composite with the highest packing fraction, and the lowest value of μ' was obtained for the 2/8 composite, which contained the highest concentration of nano ZF. The value of μ'' gradually decreased in the frequency range from 1 to 18 GHz. However, μ'' of the 4/6 composite showed an increased peak intensity at 14 GHz; this peak coincided with the first RL peak of the 4/6 composite shown as in Fig. 7 (e).

The effective bandwidth, which is the width at RL = –10 dB of the absorption peak, is the second crucial factor for evaluating the microwave absorption properties [31]. Therefore, RL under –10 dB should be considered as the effective range for microwave absorption. In Fig. 7, the RL value representing the y-axis exhibits 90% absorption ability at –10 dB, 99% at –20 dB, and 99.9% at –30 dB [32]. The NZF/ZF composites mixed with a nonmagnetic nano particle exhibited relatively good absorption properties except the sample with 10/0 mixing ratio at a thickness of 2.0 mm except the sample with 10/0 mixing ratio. The packing fraction increased for the 8/2, 6/4, and 4/6 ratio composites, the improvement rate was significant. RL was –14 dB (90% absorption) at 12.4 GHz for 8/2, –19.7 dB (99% absorption) at 12.7 GHz for 6/4, and –26 dB (99.9% absorption) at 14.2 GHz for 4/6. The packing fraction decreased for the 2/8, and 0/10 ratio composites. The absorption frequency of these composites moved to the 11 GHz range, and the absorption ability also decreased.

The permittivity and permeability of the composites were affected by the content of nano ZF. A significant increase in the permeability of the composite was observed when a large amount of nano ZF was added to it; a similar result was obtained for a large amount of NZF addition. However, the calculated RL value showed a strong correlation with the packing fraction of the composite. The data of packing fraction, magnetic properties, and effective bandwidth of the NZF/ZF composite were organized on Table 2.

4. Conclusion

In this study, we investigated the RL property changes according to the NZF packing fraction by changing the addition ratio of the ZF nanopowders. The highest packing fraction was obtained at the 4/6

mixing ratio. Moreover, as the packing fraction increased, the EM absorption frequency band shifted to a high frequency and the absorption ability improved. However, for the 2/8 and 0/10 ratio composites the absorption ability tended to decrease even for high packing fractions. The EM absorption frequency and absorption ability can be controlled using nanofillers, and a shielding composite suitable for use can be designed by changing the packing fraction of the employed nanopowder.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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